BOM KidBright Net 2 GateWay V1.0 20250522

Item	Designator	Value	Package	'Description	Part Number	Manufacturer	Qty
1	BT1	Holder2032	Hold Vertical	Battery Holder Vertical Type for CR2025,CR2032	BS-5	GLD	1
2	BZ1	KW12M85A05	PCB Hold	Magnetic Buzzer,5Vp-p/50mA(max),Dia. 12mm, Magnetic Buzzer,5Vp-p/50mA(max),Dia. 12mm, Square Wave/10cm,Coil Resistance 45Ω±15%	KW12M85A05	KINGWEI	1
3	C1, C2, C3	1nF	0603[1608]	CHIP CAP 1nF/100V, +/-10%	06031C102KAT2A	AVX Corporation	3
4	C4, C5	1uF/10V	0603[1608]	Chip Cap 1µF/10Vdc, ±10%, X7R	0603B105K100CT	WALSIN	2
5	C6	47uF/10V	0805[2012]	Chip Monolithic Ceramic Cap 47µF/10Vdc, ±20%, X5R	GRM21BR61A476ME 15L	MURATA	1
6	C7, C9, C15, C16, C17, C22	0.1uF	0603[1608]	Multilayer Ceramic Chip Cap 0.1µF/50Vdc ±10%, X7R	CL10B104KB8NFNC	SAMSUNG	6
7	C8, C18, C21	10uF/25V	0805[2012]	Multilayer Ceramic Chip Cap 10µF/25Vdc, ±10%X5R	CL21A106KAYNNNE	SAMSUNG	3
8	C10, C11	10pF	0603[1608]	Multilayer Ceramic Capacitor 10pF/50Vdc ±5% COG	CL10C100JB8NNNC	SAMSUNG	2
9	C12, C13, C14, C19, C20	0.47uF/16V	0603[1608]	MURATA MURATA, MURATA, MURATA, MURATA MURATA, MURATA	GRM188R71C474KA 88D	MURATA	5
10	D1	1N4148W-7-F	SOD-123	Fast Switching Diode 100V/300mA(Forward Continuous Fast Switching Diode 100V/300mA(Forward Continuous	1N4148W-7-F	DIODES INC.	1
11	F1	2A	1206[3216]	SMD Fuse (High Inrush Current Withstand) 2A/63V, Size 3.2mm x 1.6mm - SMD 1206, Marking N	12H1200C	LANSON	1
13	J0-1, J0-2	13 Pin Female	Pitch 2.54mm	Female Header 13 Pins, Single Row, Pitch 2.54mm, Straight Type, Plastic Height 8.5mm. Gold Plated	2212S-13G-85	NELTRON	2
14	J1	5x2	5x2 2.54mm.	Pin Header Connector 10 Pins, Right Angle Type,Double Row (5x2 Pins), Pitch 2.54mm, Gold Plated	2213R-10G	NELTRON	1
15	J3	Header 4 Female	2.54mm Pitch	P.C.Board Socket 1x4Pins Female Header Single Row, Pitch 2.54mm	FHSS04G10/RH	GTK	1
16	LD1	Blue	1206 (3216)	CHIP LED,Blue Color 160-350mcd Water Clear Lens	HL-PC-3216H203BC	HONGLITRONIC	1
17	LD2, LD3	RED	1206 (3216)	HL-PC-3216S9AC Water Clear Lens,Viewing Angle 140 Degree	HL-PC-3216S9AC	HONGLITRONIC	2
18	LD4	Amber	1206 (3216)	OASISTEK Water Clear Lens, Luminous Intensity 66 to 190mcd, Water Clear Lens, Luminous Intensity 66 to 190mcd,	TO-3216BC-MAE-NV	OASISTEK	1
19	LD5	Green	1206 (3216)	HONGLITRONIC HONGLITRONIC	HL-PC-3216U51GC	HONGLITRONIC	1
20	LD6, LD7	White	1206 (3216)	HONGLITRONIC HONGLITRONIC Yellow Diffused Lens,Viewing Angle 140 Degree Yellow Diffused Lens,Viewing Angle 140 Degree	HL-PC-3216H233W	HONGLITRONIC	2
22	Q0-1, Q0-2	BC847B,215	SOT-23	Transistor NPN 45V/0.1A, Low Current Low Voltage General Purpose Switching and Amplification	BC847B,215	NXP	2
23	Q3	BC847B,215	SOT-23	Transistor NPN 45V/0.1A, Low Current Low Voltage General Purpose Switching and Amplification	BC847B,215	NXP	1
24	Q4, Q5	2N7002LT1G	SOT-23-3	Small Signal MOSFET N-Channel 60V/115mA,225mW RDS(on) 7.5Ω Max	2N7002LT1G	ON SEMICONDU	2
25	R1, R2, R3, R26	330R	0603[1608]	RES SMD 330 OHM 5% 1/10W 0603	0603WAJ0331T5E	ROYALOHM	4
26	R4, R23, R25	1.2K	0603[1608]	Thick Film Chip Resistors 1/10W,1.2KΩ ±5%	MCR03ERTJ122	ROHM	3
27	R5, R6, R7	220R	0603[1608]	RES SMD 220 OHM 5% 1/10W 0603	0603WAJ0221T5E	ROYALOHM	3
28	R8, R11, R12, R15, R16, R19, R21, R22, R33, R34	10K	0603[1608]	Thick Film Chip Resistors 10KΩ ±1%, 1/10W	0603WAF1002T5E	ROYALOHM	10
29	R9, R32, R37	10R	0603[1608]	RES SMD 10 OHM 5% 1/10W 0603	0603WAJ0100T5E	ROYALOHM	3
30	R10	1K	0603[1608]	RES SMD 1K OHM 1% 1/10W 0603	0603WAJ0102T5E	ROYALOHM	1
31	R30, R31, R35, R36	0R	1608[0603]	RES SMD 0 OHM JUMPER 1/10W 0603	0603WAJ0000T5E	ROYALOHM	4
32	RQ1, RQ2	12K	0603[1608]	RES SMD 12K OHM 1% 1/10W 0603	MCR03ERTF1202	ROYALOHM	2
33	SW1, SW2	Tact Switch 12x12x12	SPST-2	Tactile Switches without Guide Pins Contact Rating 12V 50mA,Operating Force 160±50gf	TC-12XF-X-A00-ROHS	WEALTH METAL	2
34	SW3	Tact Switch 6x6	SPST	Tactile Switches Tactile Switches	TC-0102-X -ROHS	WEALTH METAL	1
35	U0	FT231XS-R	SSOP-28	USB to Full Handshake UART IC, USB 2.0 Compliant,UART Signal Inversion Option,Transmit and Receive LED Drive Signals.	FT232RL	FTDI	1
38	U2-1, U3-1	Header 3 Female	2.54 mm Pitch	P.C.Board Socket 1x3Pins Female Header Single Row, Pitch 2.54mm	FHSS03G10/RH	GTK	2
39	U2-2, U3-2	Header 4 Female	2.54mm Pitch	P.C.Board Socket 1x4Pins Female Header Single Row, Pitch 2.54mm	FHSS04G10/RH	GTK	2
40	U4, U7, U8	BU33SD5WG-TR	SOT23-5	CMOS LDO Regulators 1Ch 3.3V/500mA Accuracy ±2%, for Portable Equipment	BU33SD5WG-TR	ROHM	3
41	U5	MCP7940NT-I/SN	SOIC-8	Battery-Backed I2C Real-Time Clock/Calendar(RTCC) with SRAM 64Bytes	MCP7940NT-I/SN	MICROCHIP	1
42	U6	PCA9555PW,118	TSSOP-24	16-Bit GPIO Expansion for I2C/SMBus Applications, 400kHz Clock Frequency,Active Low Interrupt Output	PCA9555PW,118	NXP/NEXPERIA/	1
43	USB0	USB Type C pcb	2.54mm.x6	USB Type-C Connector Board DIY 3.1 Serial Basic Breakout	Module USB C	Pcbfun	1
44	USB0-1	Header 6 male	2.54mm.x6	Pin Header Connector 6 Pins, Straight Type, Single Row, Pitch 2.54mm, Gold Plated	2211S-06G	NELTRON	1
45	Y1	32.768KHz	SMD	Crystal Unit, FC-135R, Frequency 32.768000kHz	X1A000141000600	EPSON	1
46	Battery CR 2032	CR2032 3V/225mAh	CR 2032	Lithium Coin Battery 3V/225mAh,Diameter 20mm	CR-2032L/BE	PANASONIC	1
47	Mini Jumper Pitch 2.54mm	Jumper pitch 2.54	Jumper	Mini Jumper Pitch 2.54mm, Type B, Gold Plated Contact, 6.47mm. Length Close Type	2228BG	NELTRON	4
48	OlED 1.3" 4 Pin	OlED 1.3" 4 Pin	OlED 1.3" 4 Pin	โมดูลหน้าจอ 1.3 นิ้ว Oled ฟ้า 128X64Spi / Iic I2C	OlED 1.3" 4 Pin	OlED 1.3" 4 Pin	1
49	usb type c data cable 2m. Hoco X14	cable usb c data 2 m.		สายชาร์จ 2m Hoco X14 Speed Charger ยาว 2 เมตร สำหรับ Type C สีดำ	usb c cable Hoco X14	usb cable data	1
50	ชุดสายชาร์จโทรศัพท์ HOCO C106 สาย Micro/Type-C/iOS 1 พอร์ต	wall charger USB-A	USB charge	ชุดสายชาร์จโทรศัพท์ HOCO C106 สาย Micro/Type-C/iOS 1 พอร์ต	HOCO C106	หัวชาร์ท	1
51	POE Splitter 5V type c	POE Splitter 5V type c	POE Splitter 5V type c	Mavlo POE Splitter 5V MicroUSB Power Over Ethernet 48V ถึง 5V POE Splitter สำหรับกล้อง IP MicroUSB ประเภท C DC5 5x2 1 DC3 5	POE Splitter 5V type c		1
52	PCB KBNet2 GateWay			PCB 100 mm. x 100 mm. FR4 1.6 mm Soder mark : red Overlay : white			1

53	Spacer M2.5 x11 mm.	Spacer M2.5x11	Female to Female M2.5 x 11	M2.5 x 11 mm. Hex Brass Motherboard Standoffs Female Spacing Screws Metric Thread Board Rack Stud Column Bolt Spacers 1 order บรรจุ 30 ตัว			7
54	Screw M2.5 x 5 mm.	Screw M2.5 x 5 mm.	M2.5 x 5	Screw M2.5 x 5 mm.			1
55	สกรูขันพลาสติก M3 x 6 mm.		M3 x 6	สกรู หัวกลม สกรู น็อต หัวกลม แฉก + ขันพลาสติก M3 x 6 mm. 1 order บรรจุ 400 ตัว			1
56	กล่องกระดาษ KidBright net			กล่องกระดาษ KidBright net LOGO ลิง			60
57	LoRa Module E28-2G4T12S			LoRa Module E28-2G4T12S			2
58	U1-1	WT32-ETH01	WT32-ETH01	WT32-ETH01 Embedded Serial Port Networking Ethernet Bluetooth-compatible WIFI Combo Gateway MCU ESP32 Wireless Module Board	WT32-ETH01	WT32-ETH01	1